

Perfect Competition

Chapter 22

The Theory of Perfect Competition

- A market structure is a firm's particular environment, the characteristics of which influence the firm's pricing and decision-making.
- Perfect Competition Theory is a theory of market structure.

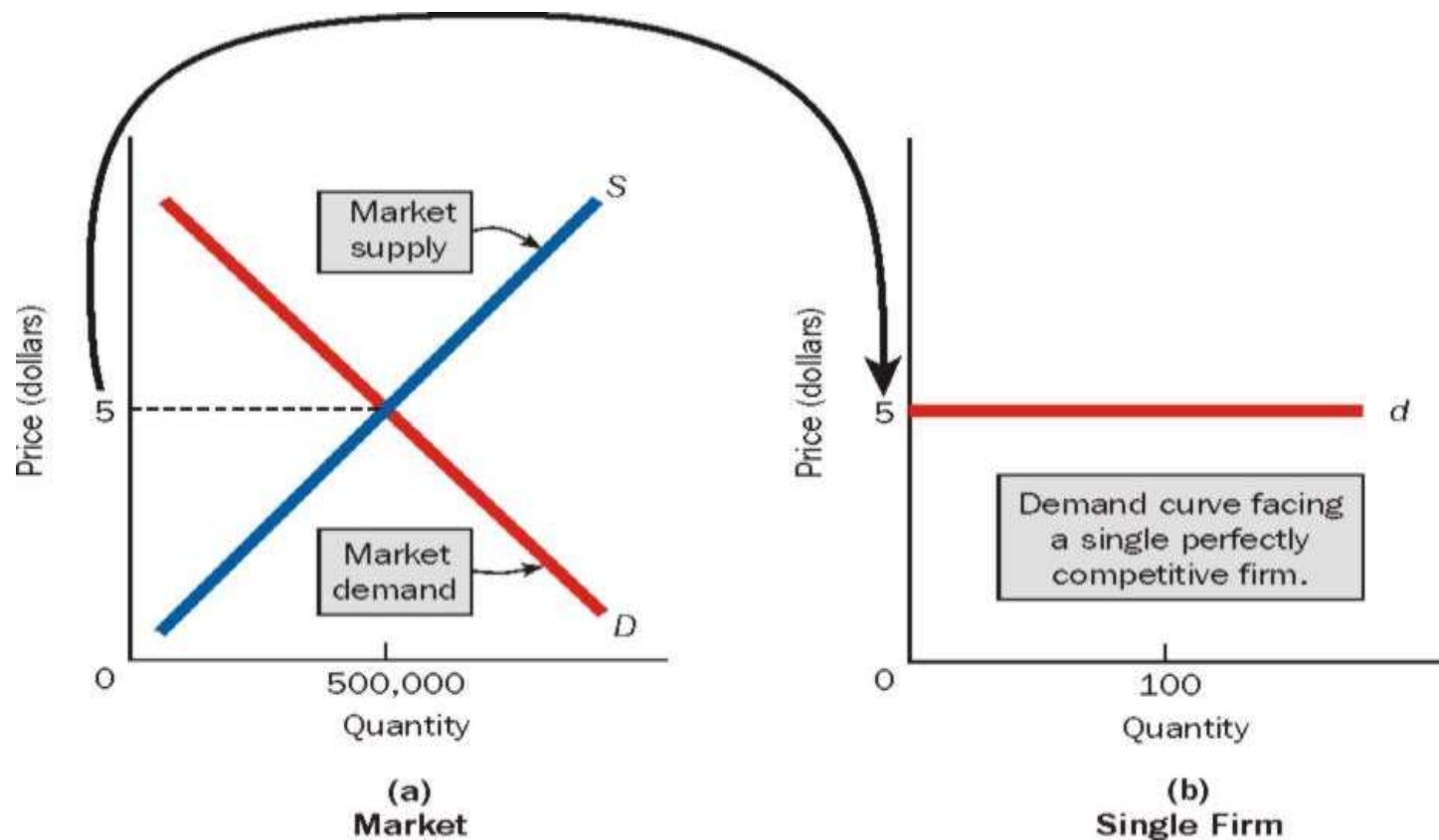
Perfect Competition Assumptions

- There are many sellers and many buyers, none of which is large in relation to total sales or purchases.
- Each firm produces and sells a homogeneous product.
- Buyers and sellers have all relevant information about prices, product quality, sources of supply, and so forth.
- Firms have easy entry and exit.

Perfectly Competitive Firms are Price Takers

- A price taker is a seller that does not have the ability to control the price of the product it sells; it takes the price determined in the market.
- A firm is restrained from being anything but a price taker if it finds itself one among many firms where its supply is small relative to the total market supply, and it sells a homogeneous product in an environment where buyers and sellers have all relevant information.

The Demand Curve for a Perfectly Competitive Firm is Horizontal!



- The market , composed of all buyers and sellers, establishes the equilibrium price.
- When the equilibrium price has been established, a single perfectly competitive firm faces a horizontal demand curve (perfectly elastic) at the equilibrium price. So, the firm is a price taker; it takes the equilibrium price established by the market and sells any and all quantities of output at this price.
- If a firm tries to charge a price higher than the equilibrium price, it won't be able to do so. This is because the firm sells homogeneous product, its supply is small relative to the total market supply, and all the buyers are well informed about where they can obtain the product at the lowest price.

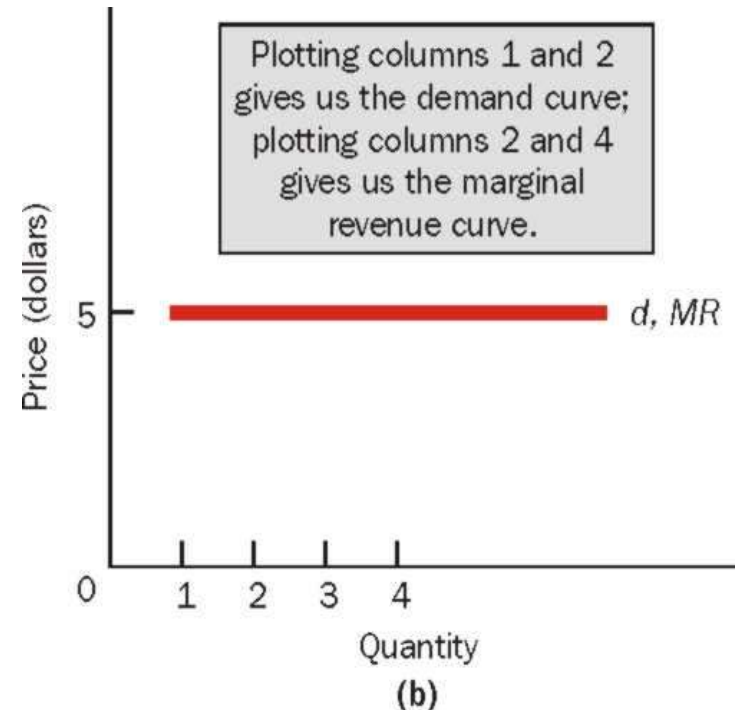
The Marginal Revenue Curve of a Perfectly Competitive Curve is the Same as its Demand Curve

- The firm's marginal revenue is the change in total revenue that results from selling one additional unit of output.
- Notice that marginal revenue at any output level is always equal to the equilibrium price. For a perfectly competitive firm, price is equal to marginal revenue.
- The marginal revenue curve for the perfectly competitive firm is the same as its demand curve.

The Demand Curve and the Marginal Revenue Curve for a Perfectly Competitive Firm

(1) Price	(2) Quantity	(3) Total Revenue = (1) $\times$ (2)	(4) Marginal Revenue = $\Delta TR / \Delta Q = \Delta(3) / \Delta(2)$
\$5	1	\$ 5	\$5
5	2	10	5
5	3	15	5
5	4	20	5

(a)



Theory and Real-world Markets

- A market that does not meet the assumptions of perfect competition may nonetheless approximate those assumptions to such a degree that it behaves as if it were a perfectly competitive market. If so, the theory of perfect competition can be used to predict the market's behavior.
- In reality, the number of sellers may not be large enough for every firm to be a price taker, but the firm's control over price may be negligible. Similarly, buyers may not have all relevant information concerning prices and quality, again it may be negligible.

Firm's decision in Perfect Competition

The perfectly competitive firm makes two decisions in the **short run**:

- Whether to produce or to shut down.
- If the decision is to produce, what quantity to produce

A firm's **long-run** decisions are:

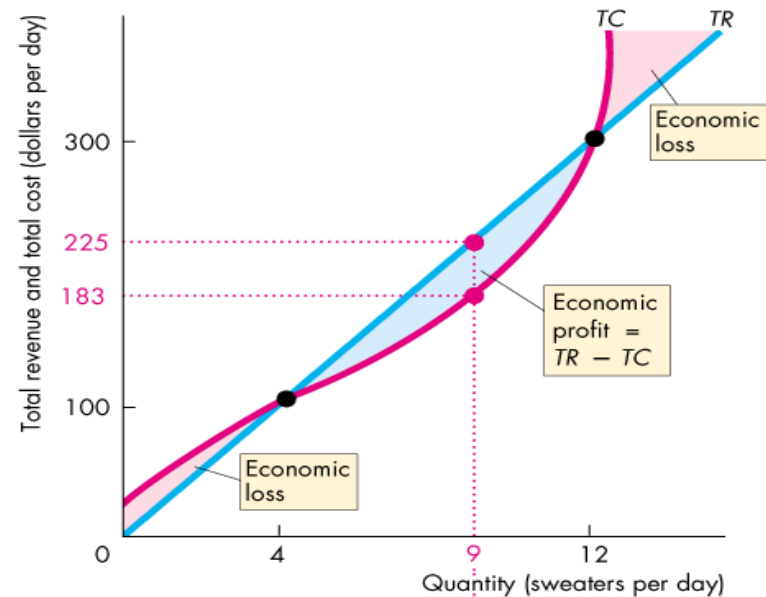
- Whether to increase or decrease its plant size.
- Whether to stay in the industry or leave it.

Short run decision

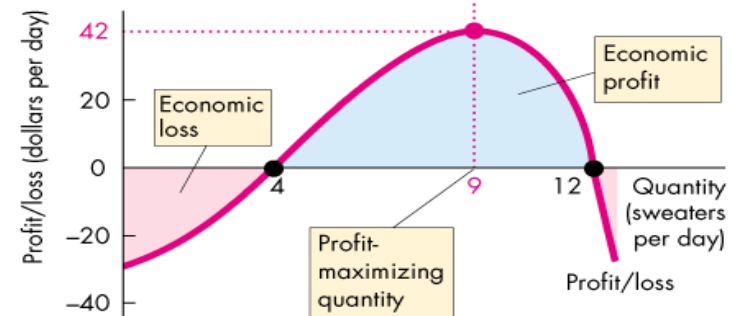
Profit Maximizing Output:

A perfectly competitive firm chooses the output that maximizes its economic profit.

One way to find the profit-maximizing output is to look at the firm's total revenue and total cost curves.



(a) Revenue and cost



(b) Economic profit and loss

- At low output levels, the firm incurs an economic loss—it can't cover its fixed costs.
- At intermediate output levels, the firm earns an economic profit
- At high output levels, the firm again incurs an economic loss—now it faces steeply rising costs because of diminishing returns.

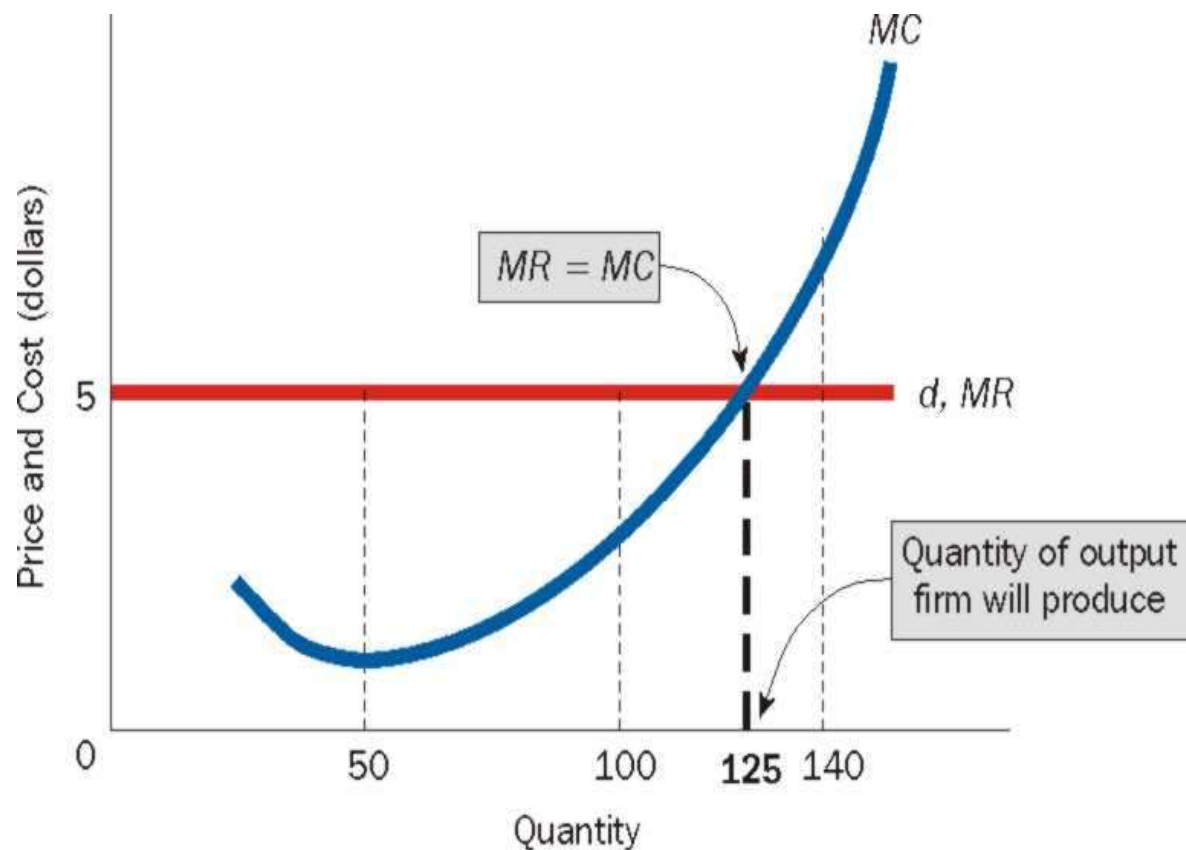
Marginal Analysis

- The firm will continue to increase its quantity of output as long as marginal revenue is greater than marginal cost. If $MR > MC$, economic profit increases if output increases.
- If $MR < MC$, economic profit decreases if output increases.
- The firm will stop increasing its quantity of output when marginal revenue and marginal cost are equal. If $MR = MC$, economic profit decreases if output changes in either direction, so economic profit is maximized.
- The Profit–Maximization Rule: Produce the quantity of output at which $MR=MC$.

In perfect competition, $P = MR = MC$

The Quantity of Output the Perfectly Competitive Firm Will Produce

The firm's demand curve is horizontal at the equilibrium price. Its demand curve is its marginal revenue curve. The firm produces that quantity of output at which $MR=MC$



Allocative efficiency and perfect competition

- Resource Allocative Efficiency: the situation occurs when produce the quantity of output at which price equals marginal cost: $P = MC$.
- Producing a good until price equals marginal cost ensures that all units of the good are produced that are of greater value to buyers than the alternative goods that might have been produced. (example)

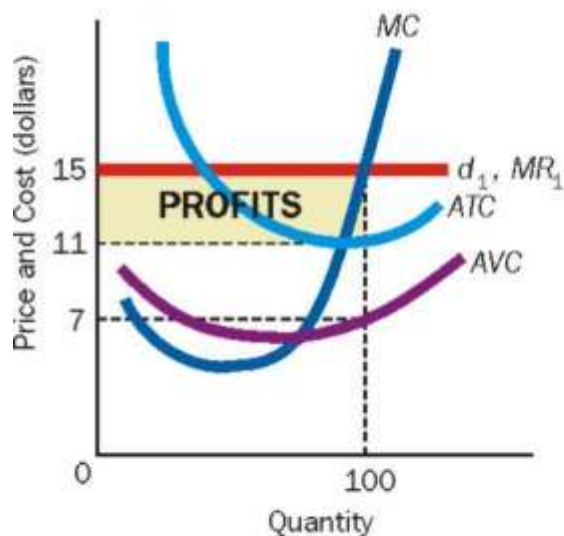
Profit Maximization and Loss Minimization for the Perfectly Competitive Firm: Three Cases

$P > ATC (> AVC)$

$TR = \$1,500$
 $TC = \$1,100$
 $TVC = \$700$
 $TFC = \$400$

Profits = \$400

Continue to produce in the short run.



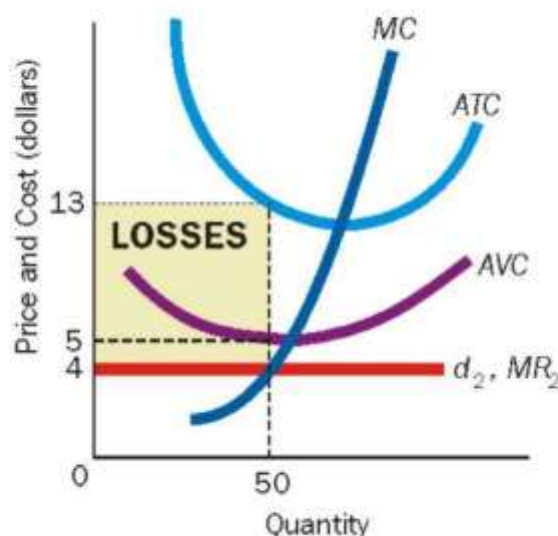
(a)
Case 1

$P < AVC (< ATC)$

$TR = \$200$
 $TC = \$650$
 $TVC = \$250$
 $TFC = \$400$

Losses = \$450

Shut down in the short run.



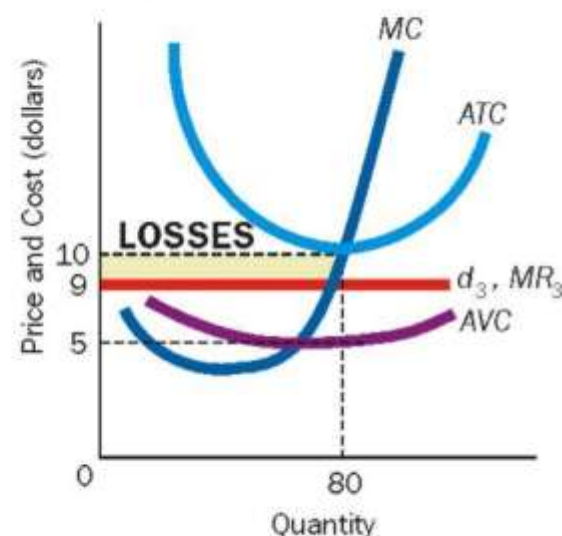
(b)
Case 2

$ATC > P > AVC$

$TR = \$720$
 $TC = \$800$
 $TVC = \$400$
 $TFC = \$400$

Losses = \$80

Continue to produce in the short run.



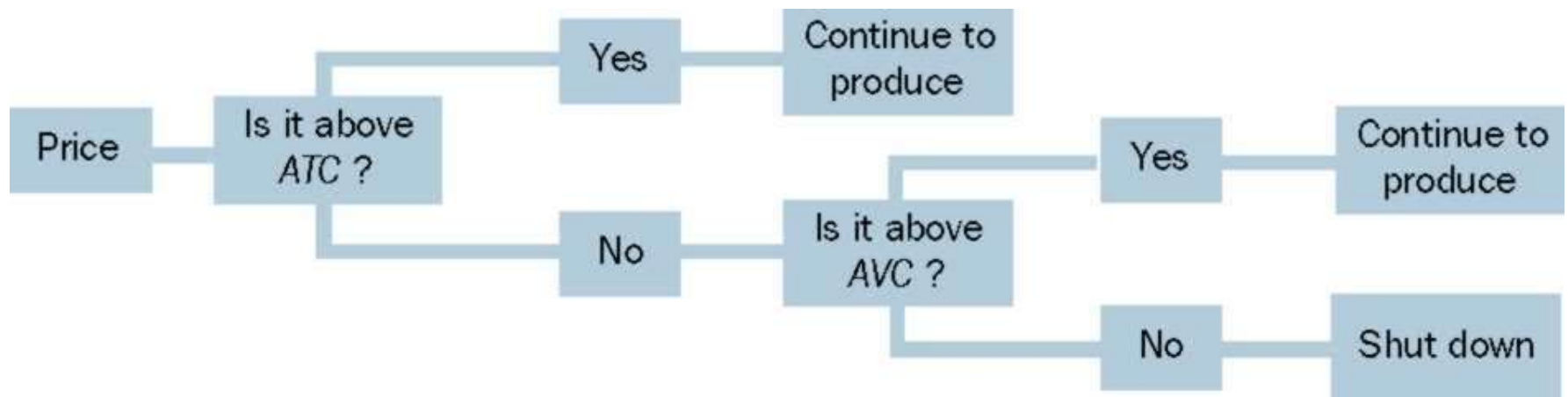
(c)
Case 3

Profit Maximization and Loss Minimization for Perfect Competition

- A firm produces in the short run as long as price is above average variable cost.
- A firm shuts down in the short run if price is less than average variable cost.
- A firm produces in the short run as long as total revenue is greater than total variable costs.
- A firm shuts down in the short run if total revenue is less than total variable costs.

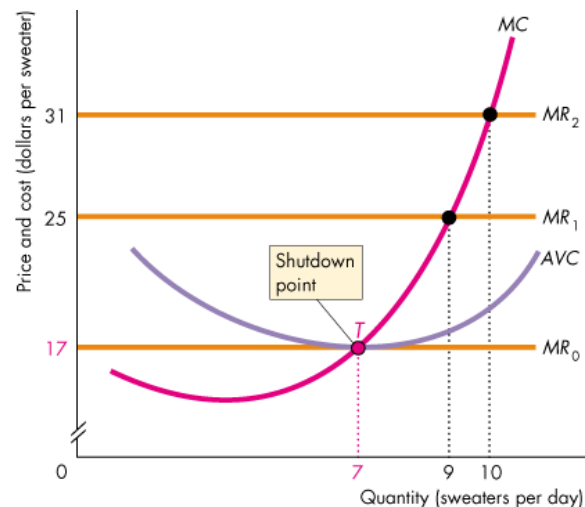
What Should a Firm Do in the Short Run?

The firm should produce in the short run as long as price (P) is above average variable cost (AVC). It should shut down in the short run if price is below average variable cost.

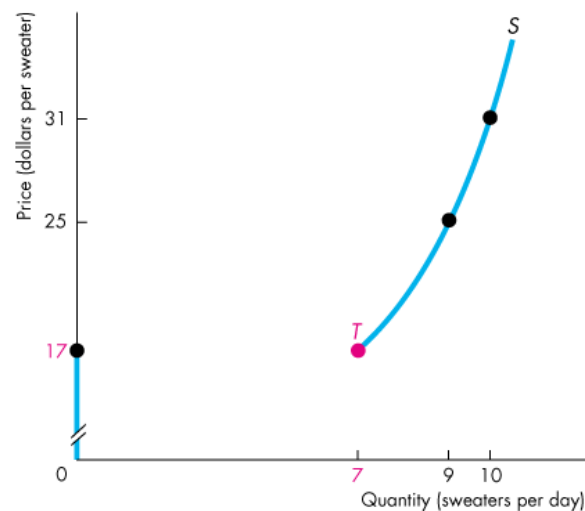


Perfectly Competitive Firm's Short-Run Supply Curve

- Only a price above average variable cost will induce the firm to supply output.
- The Short-Run supply curve is that portion of the firm's marginal cost curve that lies above the average variable cost curve.

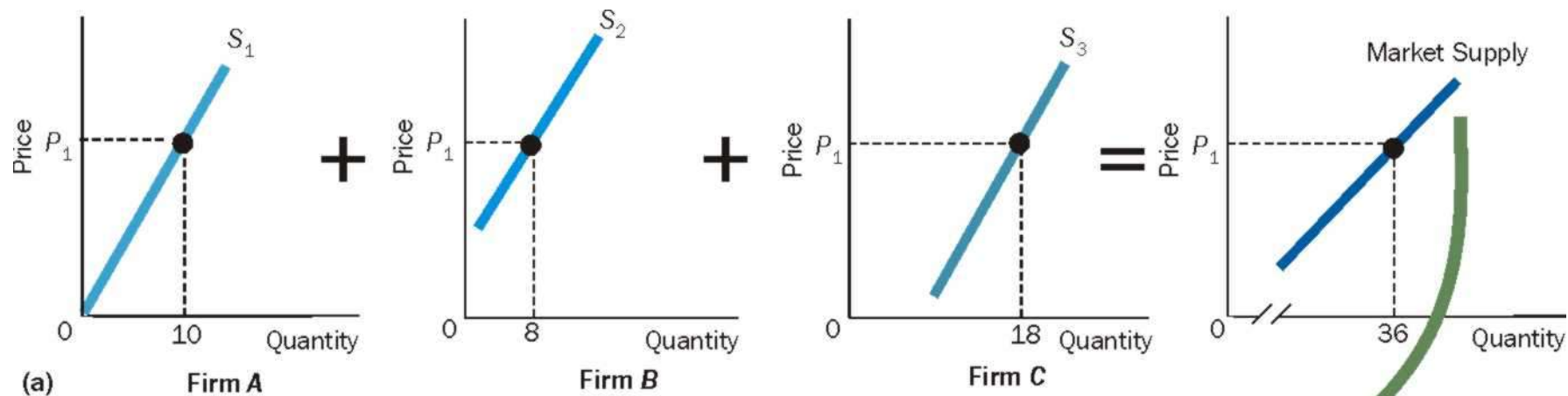


(a) Marginal cost and average variable cost



(b) Cindy's short-run supply curve

Deriving the Market (Industry) Supply Curve



Perfect Competition In The Long Run

The following conditions characterize long run equilibrium:

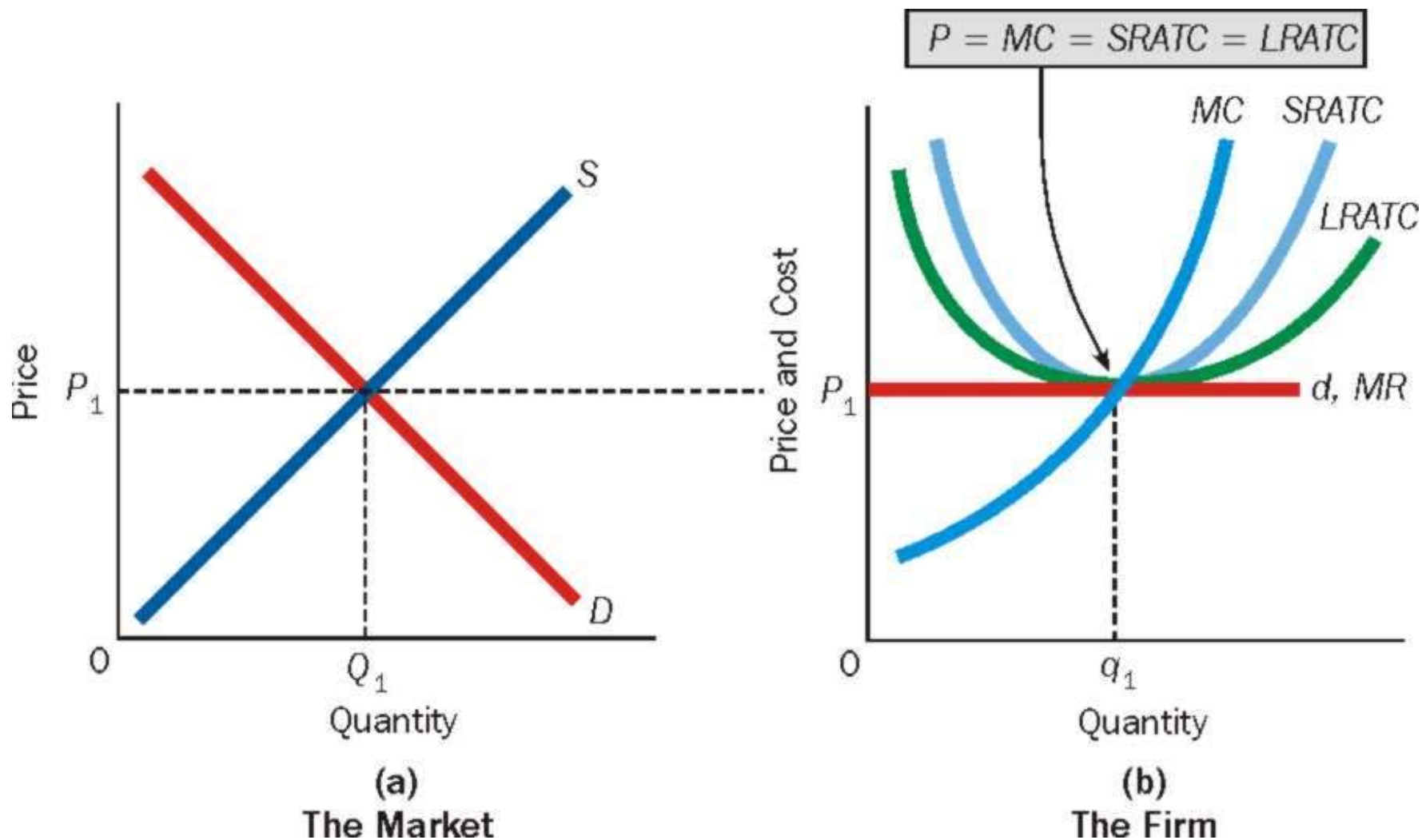
1. Economic profit is Zero: Price is equal to short-run average total cost (*SRATC*).

If $P > SRATC \rightarrow$ positive economic profit $\rightarrow$ entry

If $P < SRATC \rightarrow$ loss $\rightarrow$ exit

2. Firms are producing the quantity of output at which Price is equal to Marginal Cost (*MC*) or $MR=MC$
3. No firm has an incentive to change its plant size to produce its current output; that is, $SRATC=LRATC$ at the quantity of output at which $P=MC$. (if $SRATC > LRATC$ then firm has an incentive to change the plant size)

Long-Run Competitive Equilibrium

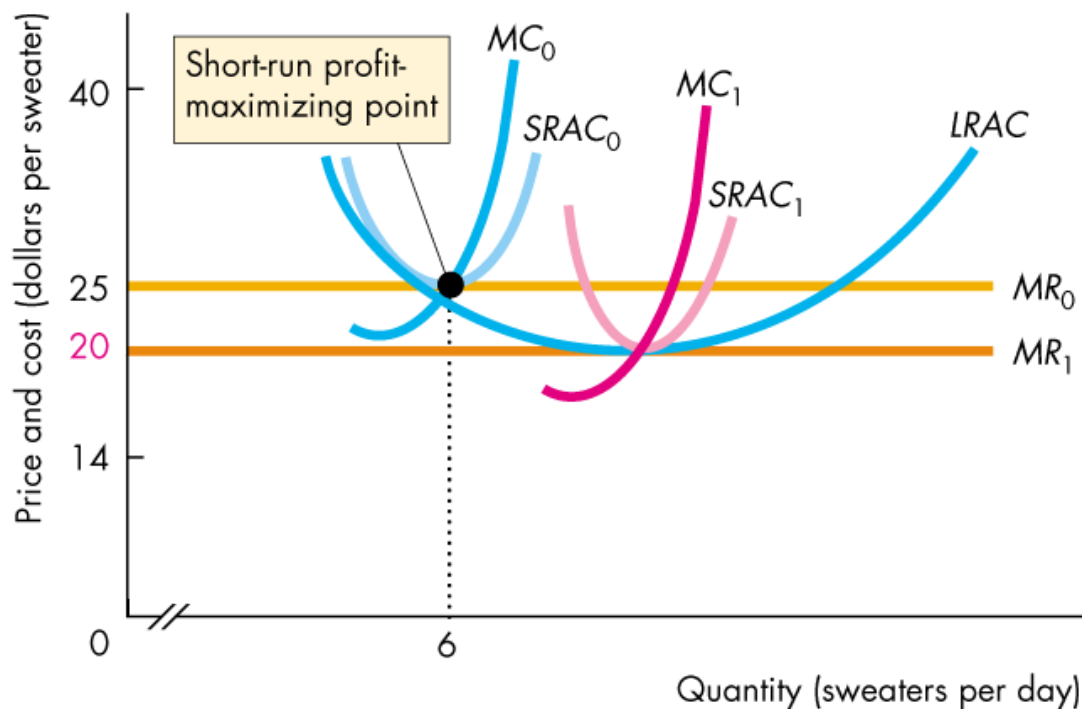


Long Run Competitive Equilibrium Exists When The Following Occur

- There is no incentive for firms to enter or exit the industry
- There is no incentive for firms to produce more or less output.
- There is no incentive for firms to change plant size. Long-run average cost is at its minimum, so firms don't change their plant size.

Condition: $P = MC = SRATC = LRATC$.

As the plant size increases, short-run supply increases, the price falls, and economic profit decreases. **Long-run equilibrium occurs when the firm is producing at the minimum long-run average cost and earning zero economic profit.**



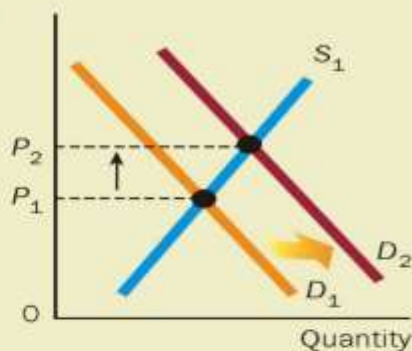
An Increase in Market Demand Throws an Industry Out of Long-Run Equilibrium

(1)

The industry is in long-run competitive equilibrium. All firms earn zero economic profit.

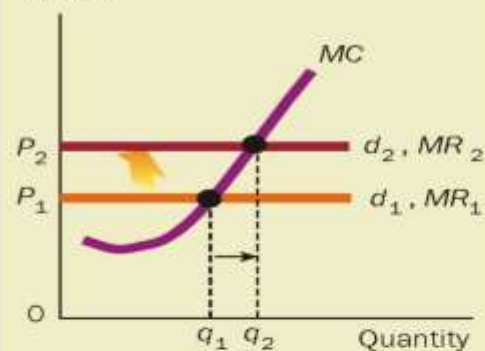
(2)

For some reason, the market demand curve rises and price rises.



(3)

This raises the demand and marginal revenue curves for the firm, and it produces more output.



(4)

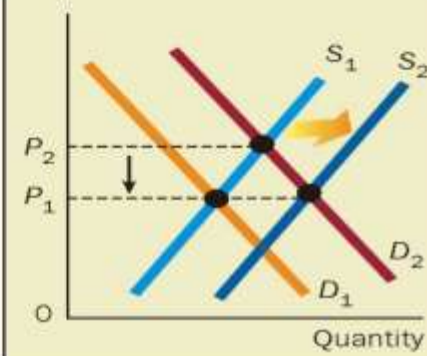
At a higher price and demand curve, firms in the industry are now earning positive economic profits.

(5)

Other firms (currently not in the industry) view the positive economic profits as an incentive to join the industry.

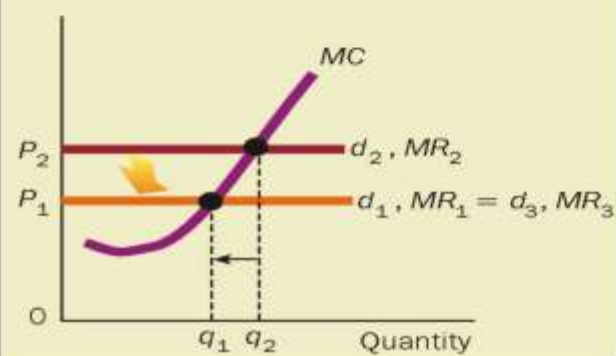
(6)

As new firms join the industry, the market supply curve shifts to the right and price declines.



(7)

This lowers the demand and marginal revenue curves for firms. Older firms, which made up the industry before market demand increased (back in Box 2), cut back output.



(8)

Eventually, all firms earn zero economic profit and are in long-run competitive equilibrium.

- Q: if price first rises owing to an increase in market demand and later falls owing to an increase in market supply, will the new equilibrium price be greater than, less than or equal to the original equilibrium price?

The answer depends whether the industry is an increasing cost or decreasing cost or constant cost industry.

Industry & Cost Relationships

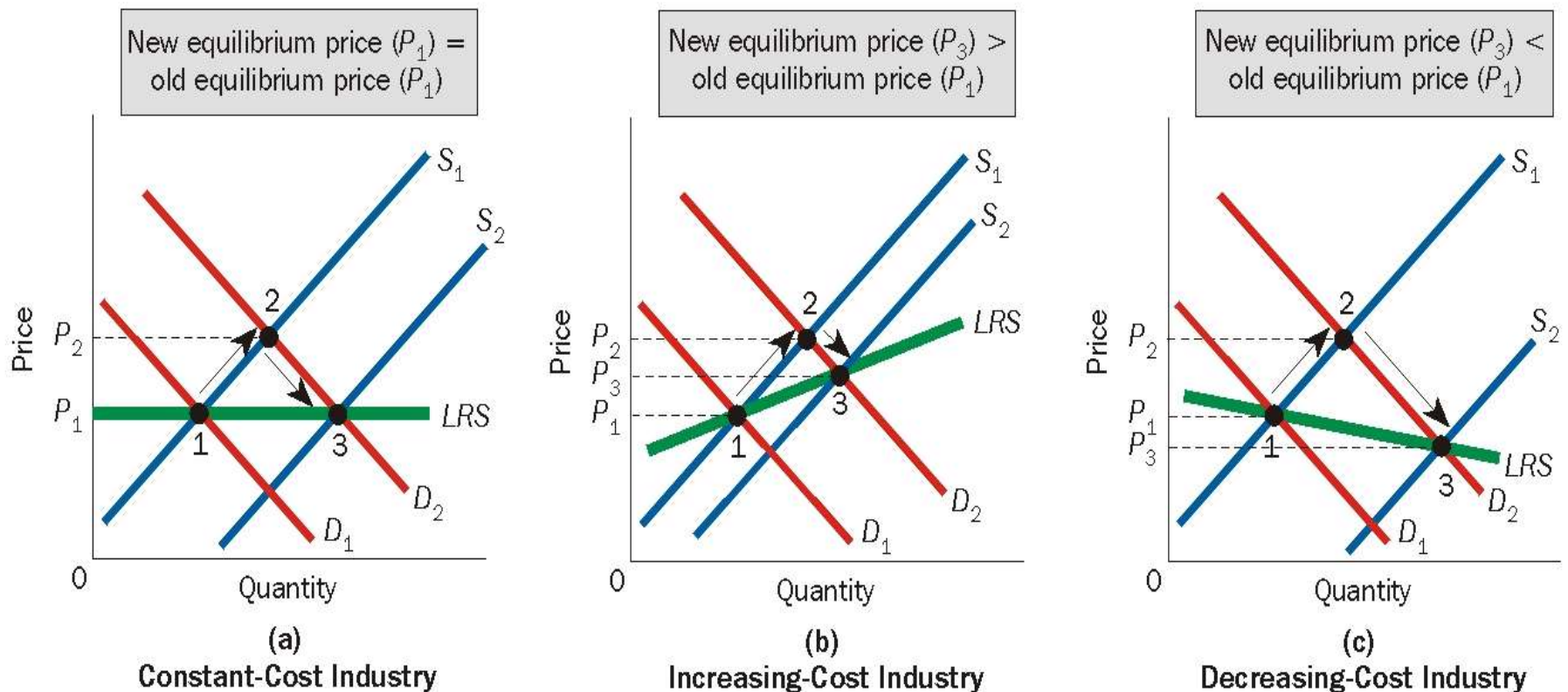
- In a **Constant-Cost Industry**, average total costs do not change as output increases or decreases when firms enter or exit the market or industry. If market demand increases for a good produced by firms in a constant cost industry, price will initially rise and then will finally fall to its original level.

for a constant cost industry, output is increased without a change in the price of inputs. Because of this, the firms' cost curves do not shift. But if costs do not rise to reduce the profits in the industry, then price must fall. Price must fall to its original level before profits can be zero, implying that the supply curve shifts rightward by the same amount that the demand curve shifts rightward.

- In an **Increasing-Cost Industry**, average total costs increase as output increases and decrease as output decreases when firms enter and exit the industry. This industry is characterized by an upward-sloping Long-run supply curve. If market demand increases for a good produced by firms in a increasing cost industry, price will initially rise and then will finally fall to a level above its original level. If there is positive economic profit firms will enter and output will increase. In an increasing cost industry, as firms purchase more inputs to produce more output, price of input increases and cost curve shifts. So profit will fall for two reasons – fall of price and increase of costs. Now as costs are rising and price is falling , then it is not necessary for price to fall to its original level. So equilibrium price is higher than the original price.

Long-Run Industry Supply Curves

In a Decreasing-Cost Industry, average total costs decrease as output increases and increases as output decreases when firms enter and exit the industry.



- In a **decreasing cost industry**, average total costs decreases as output increases as firms enter the industry, and they increase as output decreases and firms exit the industry.

In such an industry, average total costs decrease as new firms enter the industry; so price must fall below its original level to eliminate profits. A decreasing cost industry is characterized by a downward sloping long run supply curve.

Industry Adjustment to A Decrease In Demand

- The analysis outlined for an increase in demand can be reversed to explain industry adjustment to a decrease in demand.
- Some firms in the industry will decrease production because marginal revenue intersects marginal cost at a lower level of output and some firms will shut down.
- In the Long Run, some firms will leave the industry because price is below average total cost and they are suffering continual losses. As firms leave the industry, the market supply shifts leftward, and the equilibrium price rises.
- The equilibrium price will rise until long-run competitive equilibrium is reestablished and at zero economic profits

Profit And Discrimination

- A firm's discriminatory behavior can affect its profits in the context of the model of perfect competition.
- If a firm is in a perfectly competitive market structure, it will pay penalties if it chooses to discriminate.
- The greater the penalties, the less discrimination there will be.

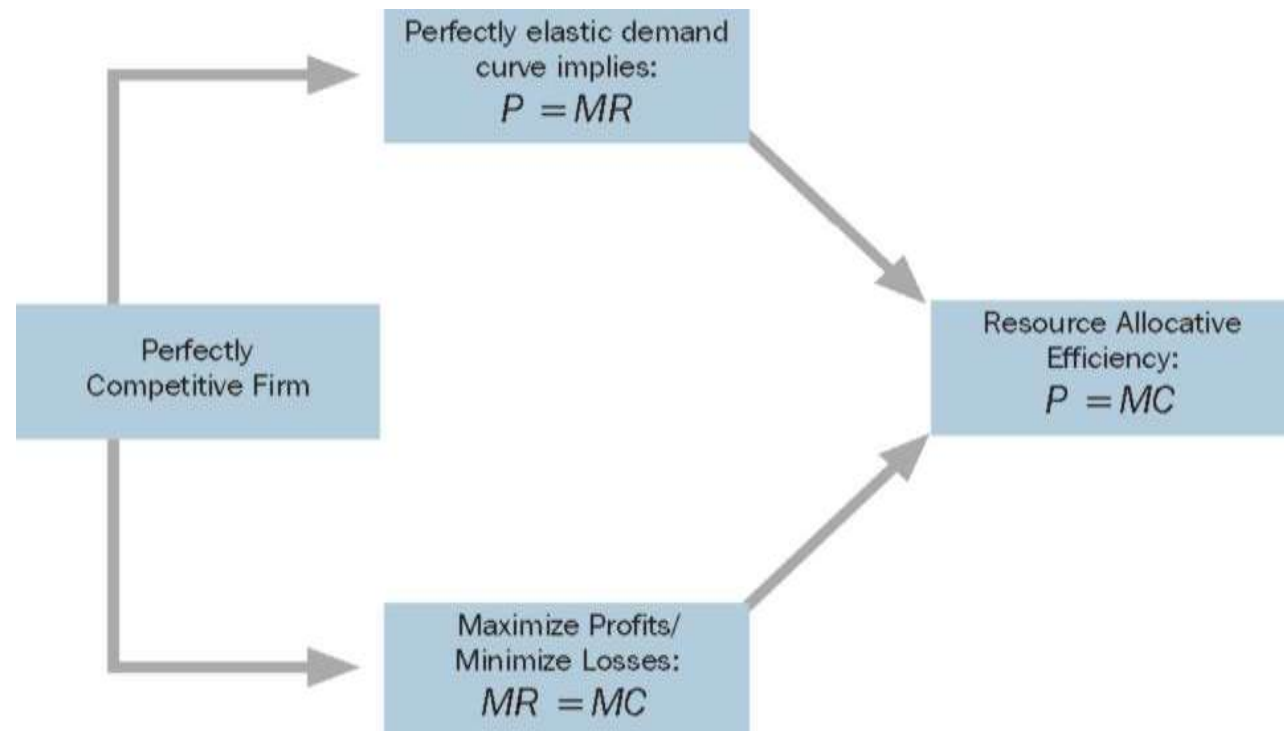
Resource Allocative Efficiency and Productive Efficiency

- A firm that produces its output at the lowest possible per unit cost is said to exhibit productive efficiency. The perfectly competitive firm does this in long run equilibrium. Productive efficiency is desirable from society's standpoint because it means that perfectly competitive firms are economizing on society's scarce resources and therefore not wasting them.
- A firm that produces the quantity of output at which $\text{Price} = \text{Marginal Cost}$ is said to exhibit resource allocative efficiency.

The Perfectly Competitive Firm and Resource Allocative Efficiency

For the perfectly competitive firm, $P=MR$. Also, the firm maximizes profits or minimizes losses by producing that quantity of output at which $MR=MC$.

Because $P=MR$ and $MR=MC$, it follows that $P=MC$, that is the perfectly competitive firm exhibits resource allocative efficiency.



Q & A

- In a perfectly competitive market, do higher costs mean higher prices?
- The perfectly competitive firm attempts to maximize profit. As a result, does it allocate resources efficiently?
- Suppose you see a product advertised on television. Does it follow that the product cannot be produced in a perfectly competitive market?